

Quiz Week 5

Groups



What is a group?

1. **Mark the correct answers. If correct, write down a short proof; if incorrect find a counter example.**

- (a) $G = \{q \in \mathbb{Q} \mid q > 0\}$ is a group with multiplication.
 (b) $G = \mathbb{R}$ with the operation $a \cdot b = a + b + 1$ is a group.
 (c) $G = \{a, b, c, d\}$ with the multiplication table given below is a group.

G	a	b	c	d
a	b	d	a	c
b	d	c	b	a
c	a	b	c	d
d	c	a	d	a

- (d) In a group, $(ab)^{-1} = a^{-1}b^{-1}$.
 (e) A group G such that $g^2 = 1$ for all $g \in G$ is abelian.
 (f) Every abelian group G satisfies $g^2 = 1$ for all $g \in G$.

Solution: (a) is correct. It is a group because it is closed under the multiplication in $\mathbb{Q}$, 1 is the identity element and the inverse of $q > 0$ is $q^{-1} \in G$.

(b) is correct. First of all notice that G is closed under $\cdot$. Secondly, the operation $\cdot$ is associative since

$$\begin{aligned}(a \cdot b) \cdot c &= (a + b + 1) \cdot c = a + b + 1 + c + 1 = a + b + c + 2, \\ a \cdot (b \cdot c) &= a \cdot (b + c + 1) = a + b + c + 1 + 1 = a + b + c + 2.\end{aligned}$$

Thus $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all a, b and c in $\mathbb{R}$. To finish check that the identity element is -1 , and the inverse of $a \in \mathbb{R}$ is $-a - 2$.

(c) is not correct. It is not a group because in the row corresponding to d the element a appears twice: $db = a$ and $d^2 = a$. Notice that the associativity laws fails since, for example:

$$(bd)d = ad = c \neq d = ba = bd^2$$

(d) is not correct. Take $G = GL_2(\mathbb{R})$ (that is, the square matrices with real coefficients and nonzero determinant) and $A = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}$. Then we have that

$$(AB)^{-1} = \frac{1}{6} \begin{pmatrix} 0 & -2 \\ 3 & 1 \end{pmatrix} \neq \frac{1}{3} \begin{pmatrix} 3 & -1 \\ 0 & 2 \end{pmatrix} = A^{-1}B^{-1}.$$

(e) is correct. In fact, let G be a group such that $g^2 = 1$ for all $g \in G$ and let us show that G is abelian. Given g_1 and g_2 in G , we must show that $g_1g_2 = g_2g_1$. To this end, apply the property of G as follows:

$$1 = (g_1g_2)^2 = g_1g_2g_1g_2 \xrightarrow{\times g_1} g_1 = g_1^2g_2g_1g_2 \xrightarrow{g_1^2=1} g_1 = g_2g_1g_2$$

as desired.

(f) is not correct. For example, $(\mathbb{R} - \{0\}, \cdot)$ is an abelian group and $2^2 = 4 \neq 1$.